PIJUSH BARAI

Barisal, Bangladesh

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PROFESSIONAL SUMMARY

I am a computer science graduate with a strong academic background and 1 years of hands-on experience implementing state-of-the-art AI architectures (e.g., CNNs, Transformers) using Python. My research experience includes developing deep learning models for practical applications such as rice disease detection. I am passionate about Machine Learning (ML), Deep Learning (DL), Natural Language Processing (NLP), and Large Language Models (LLMs). My goal is to strengthen my research expertise and become an AI professional with deep specialization in NLP and LLMs research.

EDUCATION

Bachelor of Science in Engineering January 2018 - September 2023
University of Rajshahi, Rajshahi
Field of Study: Computer Science and Engineering.
CGPA: 3.44/4.00

Higher Secondary Certificate, Class XII January 2015 - July 2017
Kabi Nazrul Govt. College, Dhaka
Field of Study: Science.
CGPA: 4.83/5.00

COURSES TAKEN (SELECTED):

- Structural Programming Language
- Statistics for Engineer
- Discrete Mathematics
- Linear Algebra
- Database Management System
- Operating System and System Programming
- Parallel Processing and Distributed System
- Cryptography and Network Security

RESEARCH EXPERIENCE

Undergraduate Thesis: Crop (Rice) Disease Detection Using Deep Learning

- Designed and implemented a deep learning–based computer vision system for automated rice disease detection using 876 leaf images
- Applied transfer learning with the InceptionResNetV2 architecture pretrained on ImageNet, incorporating custom dense layers for multi-class classification
- Collected and merged datasets from multiple public sources covering four classes (Healthy, Tungro, Sheath Blight, and Brown Spot), and performed data preprocessing, normalization, and augmentation techniques including rotation, zoom, and shear
- Evaluated the model using accuracy, confusion matrix, precision, recall, and F1-score, achieving 96% test accuracy on unseen data

RESEARCH INTERESTS

Machine Learning	Supervised, Unsupervised and Self-supervised Learning
Deep Learning	Convolutional Neural Network, Recurrent Neural Network
Natural Language Processing	Large Language Models, Text Understanding and Generation
Computer Vision	Image Classification, Object Detection, Medical Image Analysis

SKILLS

Online Judge Achievements	**LeetCode** (handle: pi129): Solved 250 problems. **Codeforces** (handle: pi129): Solved over 300 problems.
Programming language	C, C++, Java, Python, basic Kotlin, PHP(Laravel)
Machine Learning and AI	Python (NumPy, pandas, Matplotlib), Keras, TensorFlow **Computer Vision & Deep Learning:** OpenCV, Digital Image Processing Transformers (Hugging Face, BERT)
Software development	Object Oriented Programming | Design Patterns | My-SQL | HTML, CSS, Bootstrap
Others	Competitive programming | Linux (Terminal Commands, Bash/Shell) | Git

TEST SCORES

Graduate Record Examinations, October 11, 2025.
Quantitative Reasoning:160
Analytical Writing: 3.0

TOEFL iBT, November 19, 2025.
Total score :89
Reading: 26 | Listening: 19 | Speaking: 21 | Writing: 23